

1. $P(A) = \frac{\text{Số trường hợp thuận lợi cho } A}{\text{Số trường hợp có thể của phép thử}}$
2. $P(A) = \frac{m(A)}{m(\Omega)}$
3. $P(A + B) = P(A) + P(B) - P(AB)$
 $P(A + B + C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$
 $P\left(\sum_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i A_j) + \sum_{i < j < k} P(A_i A_j A_k) - \dots + (-1)^{k-1} \sum_{i_1 < i_2 < \dots < i_k} P(A_{i_1} A_{i_2} \dots A_{i_k}) + \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n)$
4. $P(A_1 + A_2 + \dots + A_n) \stackrel{?}{=} P(A_1) + P(A_2) + \dots + P(A_n)$
5. $P(A) + P(\overline{A}) = 1$
6. $P(A | B) = \frac{P(AB)}{P(B)}$
7. $P(A | B) = \frac{\text{Số trường hợp thuận lợi cho } A \text{ khi } B \text{ đã xảy ra}}{\text{Số trường hợp có thể của phép thử (tạo ra } A) \text{ khi } B \text{ đã xảy ra}}$
8. $P(A_1 A_2 \dots A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 A_2) \dots P(A_n | A_1 A_2 \dots A_{n-1})$
9. $P(AB) \stackrel{?}{=} P(A) \cdot P(B)$
10. $P(A) = \sum_{i=1}^n P(H_i) \cdot P(A | H_i)$
11. $P(H_i | A) = \frac{P(H_i) \cdot P(A | H_i)}{P(A)}$
12. $f(x) = P(X = x)$
13. $\sum_x f(x) = 1, \quad \int_{-\infty}^{\infty} f(x) dx = 1$
14. $P(X \in A) = \sum_{x \in A} f(x), \quad P(a \leq X \leq b) = \int_a^b f(x) dx$
15. $F(x) = P(X \leq x) = \begin{cases} \sum_{t \leq x} f(t) \\ \int_{-\infty}^x f(t) dt \end{cases}$
16. $F'(x) = f(x)$
 $P(a < X \leq b) = F(b) - F(a)$
17. $Y = \varphi(X): \text{Im} Y = \varphi(\text{Im} X) = \{y = \varphi(x) | x \in \text{Im} X\}; \quad P(Y = y) = P[\varphi(X) = y] = \sum_{\varphi(x)=y} f(x)$
18. $Y = \varphi(X): \quad \varphi(x) < y \Leftrightarrow a(y) < x < b(y); \quad g(y) = b'(y)f[b(y)] - a'(y)f[a(y)]$
19. $EX = \begin{cases} \sum_x x f(x) \\ \int_{-\infty}^{\infty} x f(x) dx \end{cases} : \quad E(X + Y) = EX + EY, \quad E(kX) = k \cdot EX$
20. $E[\varphi(X)] = \begin{cases} \sum_x \varphi(x) f(x) \\ \int_{-\infty}^{\infty} \varphi(x) f(x) dx \end{cases}$
21. $E(X^2) = \begin{cases} \sum_x x^2 f(x) \\ \int_{-\infty}^{\infty} x^2 f(x) dx \end{cases}$
22. $DX = E(X^2) - (EX)^2: \quad D(kX) = k^2 DX, \quad D(X + Y) \stackrel{?}{=} DX + DY$
23. $\sigma(X) = \sqrt{DX}$

24. $X \sim B(n, p): f(x) = C_n^x p^x (1-p)^{n-x}, x = 0, 1, \dots, n; EX = np, DX = np(1-p)$

25. $X \in \text{Poisson}(\lambda): f(x) = e^{-\lambda} \frac{\lambda^x}{x!}, x = 0, 1, 2, \dots; EX = \lambda, DX = \lambda$

26. $X \in U(a, b): f(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & x \notin [a, b] \end{cases}; EX = \frac{a+b}{2}, DX = \frac{(b-a)^2}{12}$

27. $X \in \text{Exp}(\lambda): f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}; EX = \frac{1}{\lambda}, DX = \frac{1}{\lambda^2}$

28. $X \in N(\mu, \sigma^2): f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; EX = \mu, DX = \sigma^2; Z = \frac{X-\mu}{\sigma} \sim N(0, 1)$

29. $f(x, y) = P(X = x, Y = y)$

30. $\sum_{x,y} f(x, y) = 1, \iint_{\mathbb{R}^2} f(x, y) dx dy = 1$

31. $P[(X, Y) \in A] = \begin{cases} \sum_{(x,y) \in A} f(x, y) \\ \iint_A f(x, y) dx dy \stackrel{?}{=} \int_a^b dx \int_{c(x)}^{d(x)} f(x, y) dy \end{cases}$

32. $F(x, y) = P(X \leq x, Y \leq y) = \begin{cases} \sum_{u \leq x, v \leq y} f(u, v) \\ \iint_{u \leq x, v \leq y} f(u, v) du dv \end{cases}$

33. $\frac{\partial^2}{\partial x \partial y} F(x, y) = f(x, y)$

34. $f_1(x) = \begin{cases} P(X = x) = \sum_y f(x, y) \\ \int_{-\infty}^{\infty} f(x, y) dy \end{cases}, f_2(y) = \begin{cases} P(Y = y) = \sum_x f(x, y) \\ \int_{-\infty}^{\infty} f(x, y) dx \end{cases}$

35. $f(x, y) \stackrel{?}{=} f_1(x) \cdot f_2(y) \quad \forall x, y$

36. $\varphi(x | Y = y) = \left[P(X = x | Y = y) = \right] \frac{f(x, y)}{f_2(y)} \quad \forall x$

37. $Z = X + Y: P(Z = z) = P(X + Y = z) = \sum_{x+y=z} f(x, y); g(z) = \int_{-\infty}^{\infty} f(x, z-x) dx$

38. $E[\varphi(X, Y)] = \begin{cases} \sum_{x,y} \varphi(x, y) \cdot f(x, y) \\ \iint_{\mathbb{R}^2} \varphi(x, y) \cdot f(x, y) dx dy \end{cases}$

39. $E(XY) = \begin{cases} \sum_{x,y} xyf(x, y) \\ \iint_{\mathbb{R}^2} xyf(x, y) dx dy \end{cases}$

40. $\text{cov}(X, Y) = E(XY) - EX \cdot EY$

41. $\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma(X) \cdot \sigma(Y)} = \frac{E(XY) - EX \cdot EY}{\sqrt{DX} \sqrt{DY}}$

42. $P(|X - EX| \geq \varepsilon) \leq \frac{DX}{\varepsilon^2}$

43. $\bar{X} \xrightarrow{p} a, \quad \frac{m}{n} \xrightarrow{p} p, \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$

44. $X \sim B(n, p), n \gg 1: P(k_1 \leq X \leq k_2) \approx \Phi\left[\frac{k_2 - np}{\sqrt{np(1-p)}}\right] - \Phi\left[\frac{k_1 - np}{\sqrt{np(1-p)}}\right]$

$$45. \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$46. \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^k m_i x_i$$

$$47. \quad S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \left(\frac{1}{n} \sum_{i=1}^n X_i^2 \right) - \bar{X}^2$$

$$48. \quad s_n^2 = \left(\frac{1}{n} \sum_{i=1}^n x_i^2 \right) - \bar{x}^2 = \left(\frac{1}{n} \sum_{i=1}^k m_i x_i^2 \right) - \bar{x}^2$$

$$49. \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n}{n-1} S_n^2$$

$$50. \quad s^2 = \frac{1}{n-1} \left[\left(\sum_{i=1}^n x_i^2 \right) - n\bar{x}^2 \right] = \frac{1}{n-1} \left[\left(\sum_{i=1}^k m_i x_i^2 \right) - n\bar{x}^2 \right]$$

$$51. \quad P(Z > z_\alpha) = \alpha, \quad P(\chi_k^2 > \chi_{\alpha,k}^2) = \alpha, \quad P(t_k > t_{\alpha,k}) = \alpha$$

52.

Tham số	Kiểu ước lượng		Ghi chú
	ƯLKC	Khoảng tin cậy với độ tin cậy $1 - \alpha$	
EX	$\bar{X}$	$DX = \sigma^2$ đã biết	với KTC: $X \sim N$
		DX chưa biết	
DX	S^2	$\left(\frac{(n-1)S^2}{\chi_{\alpha/2,n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2,n-1}^2} \right)$	
$p = P(A)$	$p^* = \frac{m}{n}$	$\left(p^* - z_{\alpha/2} \sqrt{\frac{p^*(1-p^*)}{n}}, p^* + z_{\alpha/2} \sqrt{\frac{p^*(1-p^*)}{n}} \right)$	
			$n \gg 1$

53.

H_0	H_1	Dấu hiệu / ĐK	Thống kê kiểm định	ĐK bác bỏ H_0
$EX = \mu_0$	$EX \neq \mu_0$	$DX = \sigma^2$ đã biết	$z = \frac{\bar{x} - \mu_0}{\sigma} \sqrt{n} (*)$	$ z > z_{\alpha/2}$
	$EX > \mu_0$			$z > z_\alpha$
	$EX < \mu_0$			$z < -z_\alpha$
như trên		DX chưa biết	$t = \frac{\bar{x} - \mu_0}{s} \sqrt{n} (**)$	$ t > t_{\alpha/2,n-1}$ $t > t_{\alpha,n-1}$ $t < -t_{\alpha,n-1}$
$p = p_0$	$p \neq p_0$ $p > p_0$ $p < p_0$	$n \gg 1$	$z = \frac{\frac{m}{n} - p_0}{\sqrt{p_0(1-p_0)}} \sqrt{n}$	như (*)
$EX_1 = EX_2$	$EX_1 \neq EX_2$ $EX_1 > EX_2$ $EX_1 < EX_2$	DX_1, DX_2 đã biết	$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{DX_1}{n_1} + \frac{DY}{n_2}}}$	như (*)
như trên		$DX_1 = DX_2$ chưa biết	$t = (1)$	như (**), thay $n - 1$ bởi $n_1 + n_2 - 2$

$p_1 = p_2$	$p_1 \neq p_2$ $p_1 > p_2$ $p_1 < p_2$	$n_1, n_2 \gg 1$	$z = (2)$	như (*)
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$$(1) = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(n_1 - 1)s_{\bar{X}_1}^2 + (n_2 - 1)s_{\bar{X}_2}^2}} \sqrt{\frac{n_1 n_2 (n_1 + n_2 - 2)}{n_1 + n_2}}$$

$$(2) = \left(\frac{m_1}{n_1} - \frac{m_2}{n_2} \right) \sqrt{\frac{n_1 n_2 (n_1 + n_2)}{(m_1 + m_2) (n_1 + n_2 - m_1 - m_2)}}.$$

54. $H_0 : P(A_i) = p_{i0} \forall i = \overline{1, k}$: (a) $e_i \stackrel{?}{=} np_{i0}$, (b) $\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$, (c) $\chi^2 \stackrel{?}{>} \chi_{\alpha, k-1}^2$

55. $H_0 : X \sim F(x)$: (a) $\text{Im}X = S_1 \cup S_2 \cup \dots \cup S_k$, (b) $p_{i0} = P(X \in S_i | H_0) = \begin{cases} \sum_{x \in S_i} f(x) \\ \int_{S_i} f(x) dx \end{cases}$, (55)

56. $H_0 : X \sim F(x, \theta_1, \theta_2, \dots, \theta_r) \Leftrightarrow H_0^* : X \sim F(x, \theta_1^*, \theta_2^*, \dots, \theta_r^*), \chi^2 \stackrel{?}{>} \chi_{\alpha, k-r-1}^2$:

$X \in ?$	θ	θ^*
$B(n, p)$	p	p^*
$\text{Poisson}(\lambda)$	λ	$\bar{X}$
$U(a, b)$	a, b	$\bar{X} - S_n \sqrt{3}, \bar{X} + S_n \sqrt{3}$
$\text{Exp}(\lambda)$	λ	$1/\bar{X}$
$N(\mu, \sigma^2)$	μ, σ^2	$\bar{X}, S_n^2$

57. $H_0 : X, Y$ độc lập: (a) $\text{Im}X = S_1 \cup \dots \cup S_r, \text{Im}Y = T_1 \cup \dots \cup T_c$; (b) $n_{i*} = \sum_j n_{ij}, n_{*j} = \sum_i n_{ij}$;

(c) $e_{ij} = \frac{n_{i*} \times n_{*j}}{n}$; (d) $\chi^2 = \sum_{i,j} \frac{(n_{ij} - e_{ij})^2}{e_{ij}}$; (e) $\chi^2 \stackrel{?}{>} \chi_{\alpha, (r-1)(c-1)}^2$

58. $r = \frac{\overline{XY} - \bar{X} \cdot \bar{Y}}{S_{n,X} S_{n,Y}}$

59. $Z = E(Y | X): X = x \Rightarrow Z = E(Y | X = x) = \begin{cases} \frac{1}{f_1(x)} \sum_y y f(x, y) \\ \frac{1}{f_1(x)} \int_{-\infty}^{\infty} y f(x, y) dy \end{cases}$

60. $E\{[Y - b_1 X - b_0]^2\} \rightarrow \min_{b_1, b_0}: b_1 = \rho \sqrt{\frac{DY}{DX}}, b_0 = EY - b_1 EX$

61. $\frac{1}{n} \sum_{i=1}^n (Y_i - b_1 X_i - b_0)^2 \rightarrow \min_{b_1, b_0}: b_1 = r \frac{S_Y}{S_X}, b_0 = \bar{Y} - b_1 \bar{X}$